

Scenario-Based Interview Questions – CONSTRAINTS

Scenario 1: Employee Registration System

Table: Employee

EmpID	EmpName	Email	PhoneNo	Department
101	Ravi	ravi@gmail.com	9876543210	IT
102	Meena	meena@gmail.com	9876543211	HR

Scenario

A company is developing an Employee Registration System. Every employee should have a unique Employee ID and Email ID. During testing, one employee tries to register using an Email ID that already exists in the database.

Interview Question

Which SQL Constraint should be implemented to prevent duplicate Email IDs, and why?
Unique

It ensures that no two rows can have the same email address.

Scenario 2: Online Shopping Application

Table: Orders

OrderID	CustomerID	ProductID	OrderDate
1001	201	P101	2026-07-01
1002	202	P102	2026-07-02

Scenario

An order can be created only if the Customer already exists in the Customer table. During testing, a developer attempts to insert an order for a Customer ID that is not available in the Customer table.

Interview Question

Which SQL Constraint should be used to enforce this business rule?
Foreign Key

Scenario 3: College Student Admission System

Table: Student

StudentID	StudentName	Age	Course
101	Arjun	20	BCA
102	Divya	19	B.Sc

Scenario

The college has decided that every student must provide their name during admission. While entering records, a staff member leaves the Student Name field empty.

Interview Question

Which SQL Constraint should be applied to ensure that every student record contains a Student Name? NOT NULL

Scenario 4: Banking Account Management

Table: BankAccount

AccountNo	CustomerName	AccountType	Balance
5001	John	Savings	25000
5002	Priya	Current	80000

Scenario

The bank allows only two account types: **Savings** and **Current**. During data entry, an employee enters the Account Type as "Personal".

Interview Question

Which SQL Constraint should be used to restrict invalid account types from being stored in the database? Check

Scenario 5: Company Asset Management

Table: Assets

AssetID	AssetName	Status	PurchaseDate
A101	Laptop	Active	2026-01-15
A102	Monitor	Active	2026-02-10

Scenario

Whenever a new asset is added to the system, if the user does not specify the Status, it should automatically be stored as '**Active**'.

Interview Question

Which SQL Constraint can automatically assign a value when no value is provided during insertion? Default